SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI 682 022

DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

Laboratory Manual

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

LIST OF EXPERIMENTS

1. Familiarization of Electrical symbols.
2. One lamp one switch.
3. Staircase wiring
4. Hospital wiring
5. Plug point wiring (one lamp and one plug point)
6. Familiarization of Circuit breakers and single phase distribution board
7. Verification of Basic Circuit Laws- (Ohms Law,KCL,KVL)
8. Rectifier circuits
9. Power measurement in RLC circuit.

SCHEMATIC DIAGRAM

The schematic diagram consists of idealized circuit elements each of which represents some property of the actual circuit. A well-drawn schematic makes it easy to understand how a circuit works and aids in troubleshooting; a poor schematic only creates confusion. By keeping a few rules and suggestions in mind, you can draw a good schematic in no more time than it takes to draw a poor one.

Rules of Drawing Circuit Diagrams

Drawing circuit diagrams is not difficult but it takes a little practice to draw neat, clear diagrams.

Follow these tips for best results:

- Make sure you use the correct symbol for each component
- Draw connecting wires as straight lines
- Put a 'blob' at each junction between wires.
- Label components such as lamps and switches
- The neutral wire should be at the top and the phase wire at the bottom.

First Aid for Electrical Accidents

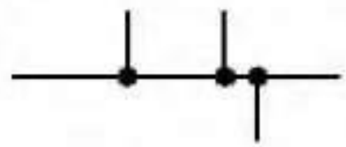
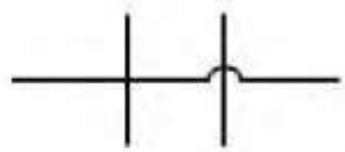
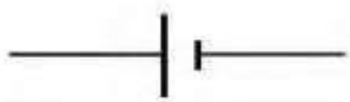
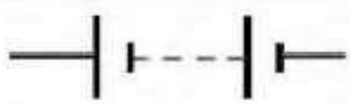
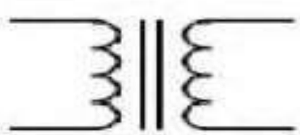
First Aid for Electric Shock Victims

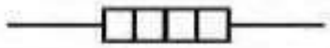
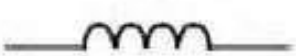
1. Don't touch them!
2. Unplug the appliance or turn off the power at the control panel.
3. If you can't turn off the power, use a piece of wood, like a broom handle, dry rope or dry clothing, to separate the victim from the power source.
4. Do not try to move a victim touching a high voltage wire. Call for emergency help.
5. Keep the victim lying down. Unconscious victims should be placed on their side to allow drainage of fluids. Do not move the victim if there is a suspicion of neck or spine injuries unless absolutely necessary.
6. If the victim is not breathing, apply mouth-to-mouth resuscitation. If the victim has no pulse, begin cardiopulmonary resuscitation (CPR). Then cover the victim with a blanket to maintain body heat, keep the victim's head low and get medical attention.

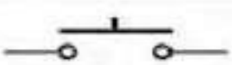
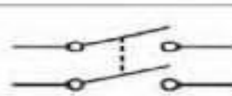
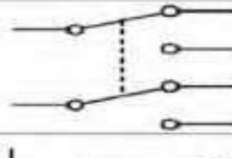
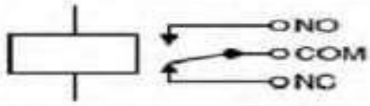
First Aid for Electrical Accidents

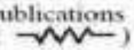
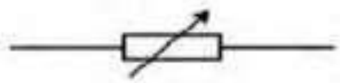
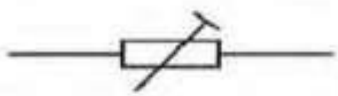
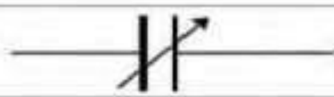
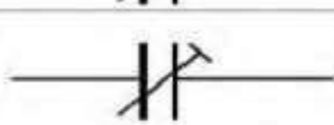
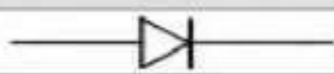
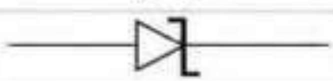
- **Disconnect the appliances or turn off the power if a person is undergoing electric shock.**
- **Cover associated electric shock burns with a dry sterile dressing only.**
- **Never touch a person undergoing electric shock or you too could become a victim.**

FAMILIARIZATION OF ELECTRICAL SYMBOLS

COMPONENT	CIRCUIT SYMBOL	DESCRIPTION
WIRE CONNECTIONS		
Wire		To pass current very easily from one part of a circuit to another.
Wires joined		A 'blob' should be drawn where wires are connected (joined), but it is sometimes omitted. Wires connected at 'crossroads' should be staggered slightly to form two T-junctions, as shown.
Wires not joined		In complex diagrams it is often necessary to draw wires crossing even though they are not connected. (prefer the 'bridge' symbol shown on the right because the simple crossing on the left may be misread as a join where you have forgotten to add a 'blob'!)
POWER SUPPLIES		
Cell		Supplies electrical energy. The larger terminal (on the left) is positive (+). A single cell is often called a battery, but strictly a battery is two or more cells joined together.
Battery		Supplies electrical energy. A battery is more than one cell. The larger terminal (on the left) is positive (+). The smaller terminal (on the right) is negative (-).
DC supply		Supplies electrical energy. DC = Direct Current, always flowing in one direction.
AC supply		Supplies electrical energy. AC = Alternating Current, continually changing direction.
Fuse		A safety device which will 'blow' (melt) if the current flowing through it exceeds a specified value.
Transformer		Two coils of wire linked by an iron core. Transformers are used to step up (increase) and step down (decrease) AC voltages. Energy is transferred between the coils by the magnetic field in the core. There is no electrical connection between the coils.
Earth (Ground)		A connection to earth. For many electronic circuits this is the 0V (zero volts) of the power supply, but for mains electricity and some radio circuits it really means the earth. It is also known as ground.

OUTPUT DEVICES: LAMPS, HEATER, MOTOR, etc.		
Lamp (lighting)		A transducer which converts electrical energy to light. This symbol is used for a lamp providing illumination, for example a car headlamp or torch bulb.
Lamp (indicator)		A transducer which converts electrical energy to light. This symbol is used for a lamp which is an indicator, for example a warning light on a car dashboard.
Heater		A transducer which converts electrical energy to heat.
Motor		A transducer which converts electrical energy to kinetic energy (motion).
Bell		A transducer which converts electrical energy to sound.
Buzzer		A transducer which converts electrical energy to sound.
Inductor (Coil, Solenoid)		A coil of wire which creates a magnetic field when current passes through it. It may have an iron core inside the coil. It can be used as a transducer converting electrical energy to mechanical energy by pulling on something.

Switches		
Push Switch (push-to-make)		A push switch allows current to flow only when the button is pressed. This is the switch used to operate a doorbell.
Push-to-Break Switch		This type of push switch is normally closed (on), it is open (off) only when the button is pressed.
On-Off Switch (SPSTS)		SPSTS = Single Pole Single Throw Switch. An on-off switch allows current to flow only when it is in the closed (on) position.
2-way Switch (SPDTS)		SPDTS = Single Pole Double Throw Switch. A 2-way changeover switch directs the flow of current to one of two routes according to its position. Some SPDT switches have a central off position and are described as 'on-off-on'.
Dual On-Off Switch (DPSTS)		DPST = Double Pole, Single Throw Switch. A dual on-off switch which is often used to switch mains electricity because it can isolate both the live and neutral connections.
Reversing Switch (DPDTS)		DPDT = Double Pole, Double Throw Switch. This switch can be wired up as a reversing switch for a motor. Some DPDT switches have a central off position.
Relay		An electrically operated switch, for example a 9V battery circuit connected to the coil can switch a 230V AC mains circuit. NO = Normally Open, COM = Common, NC = Normally Closed.

Resistors		
Resistor		A resistor restricts the flow of current, for example to limit the current passing through an LED. A resistor is used with a capacitor in a timing circuit. (Some publications also use the old resistor symbol: )
Variable Resistor (Rheostat)		This type of variable resistor with 2 contacts (a rheostat) is usually used to control current. Examples include: adjusting lamp brightness, adjusting motor speed, and adjusting the rate of flow of charge into a capacitor in a timing circuit.
Variable Resistor (Potentiometer)		This type of variable resistor with 3 contacts (a potentiometer) is usually used to control voltage. It can be used like this as a transducer converting position (angle of the control spindle) to an electrical signal.
Variable Resistor (Preset)		This type of variable resistor (a preset) is operated with a small screwdriver or similar tool. It is designed to be set when the circuit is made and then left without further adjustment. Presets are cheaper than normal variable resistors so they are often used in projects to reduce the cost.
CAPACITORS		
Capacitor		A capacitor stores electric charge. A capacitor is used with a resistor in a timing circuit. It can also be used as a filter, to block DC signals but pass AC signals.
Capacitor polarized		A capacitor stores electric charge. This type must be connected the correct way round. A capacitor is used with a resistor in a timing circuit. It can also be used as a filter, to block DC signals but pass AC signals.
Variable Capacitor		A variable capacitor is used in a radio tuner.
Trimmer Capacitor		This type of variable capacitor (a trimmer) is operated with a small screwdriver or similar tool. It is designed to be set when the circuit is made and then left without further adjustment.
DIODES		
Diode		A device which only allows current to flow in one direction.
LED Light Emitting Diode		A transducer which converts electrical energy to light.
Zener Diode		A special diode which is used to maintain a fixed voltage across its terminals.

EXERCISES

ONE LAMP CONTROLLED BY ONE SWITCH

AIM: One lamp controlled by one switch

Design and wire up a circuit in conduit system of wiring as per the given layout (also prepare an estimate of materials required), which satisfy the following conditions

DESIGN

Power (load) =

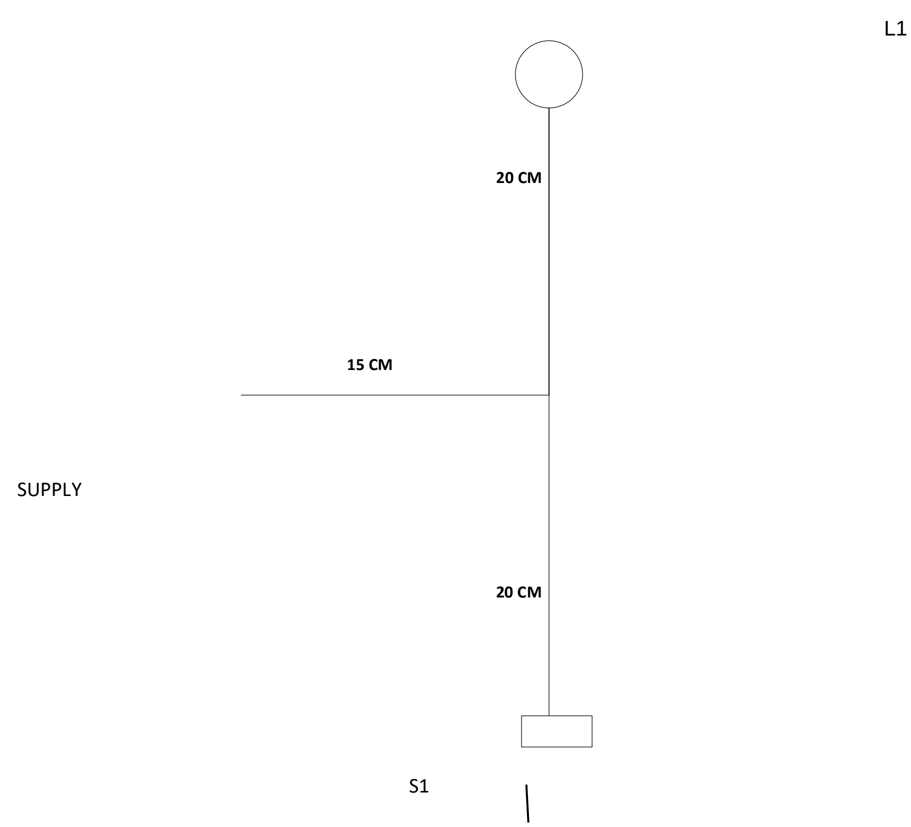
Power P =VI COSϕ

Load current

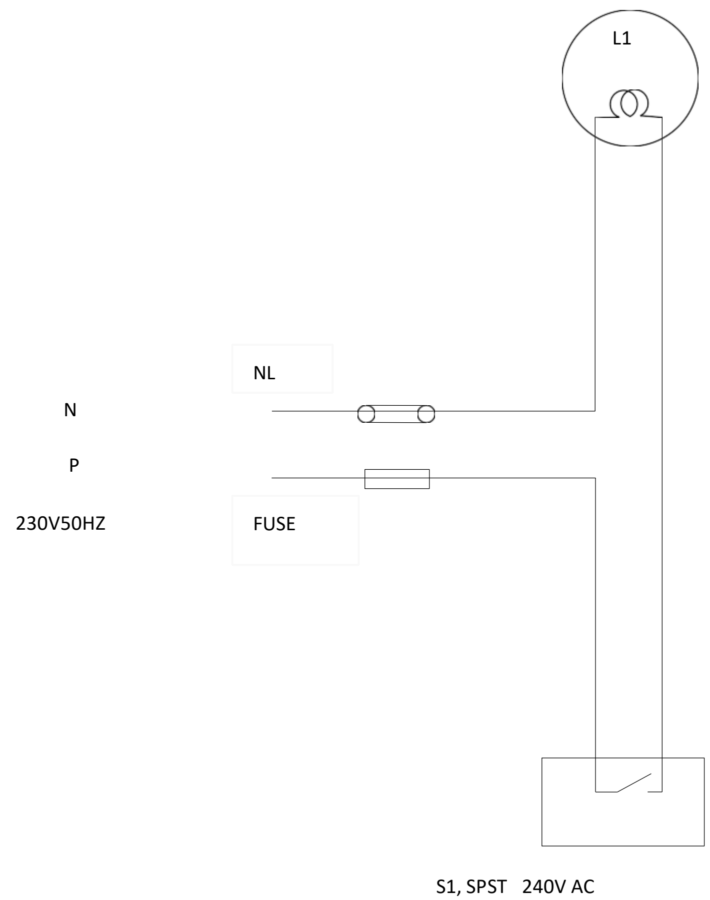
Power/Voltage =

Fuse wire rating =1.5 x Load current

LAYOUT DIAGRAM



CIRCUIT DIAGRAM



CONDITIONS

SWITCH	**LAMP**
OFF	DARK
ON	BRIGHT

MATERIALS REQUIRED:

SL NO	ITEMS	SPECIFICATIONS	RATING	QUANTITY
1	Conduit pipe	20mm PVC		
2	PVC insulated copper wire	$1mm^2$	5A	
3	Fuse unit	Kitkat (ordinary type)	16 A,250V	
4	Neutral link	Porcelain base with copper segment	16 A,250V	
5	Saddles	20mm Metal		
6	Switch	SPST/SPDT	5A ⁄250V	
7	Round block	3inch PVC		
8	Holder	Study batten	5A ⁄250V	
9	20mm PVC Elbow or Junction boxes	20mm PVC(3 WAY,4 WAY)		
10	Switch boxes	3inch x 2inch PVC		
11	lamp	ordinary type incandescent	60W	
12	Screws	½ inch ¾ inch 1 ¼ inch		

TOOLS REQUIRED:

Screw driver, working board, pocketknife, pliers, pocker, hacksaw, connector, chisel, mallet, steel rule, etc

RESULT:

The wiring has carried out successfully as per the given layout and conditions

STAIRCASE WIRING

AIM: Staircase wiring

Design and wire up a circuit in conduit system of wiring as per the given layout (also prepare an estimate of materials required), which satisfy the following conditions

DESIGN

Power (load) =

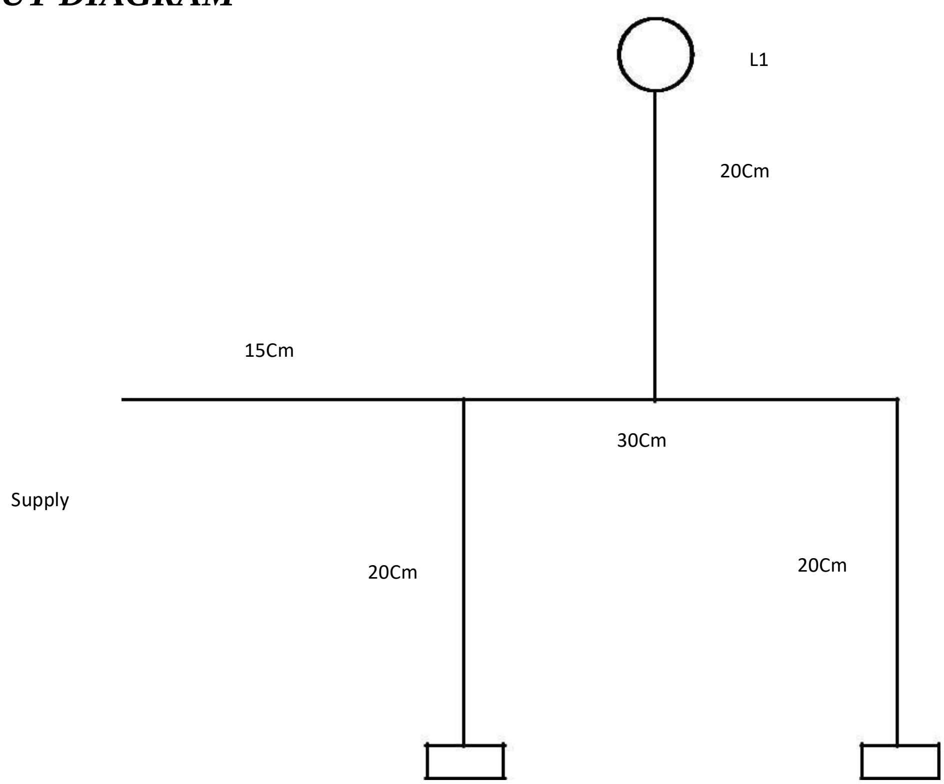
Power P =VI $COS\phi$

Load current

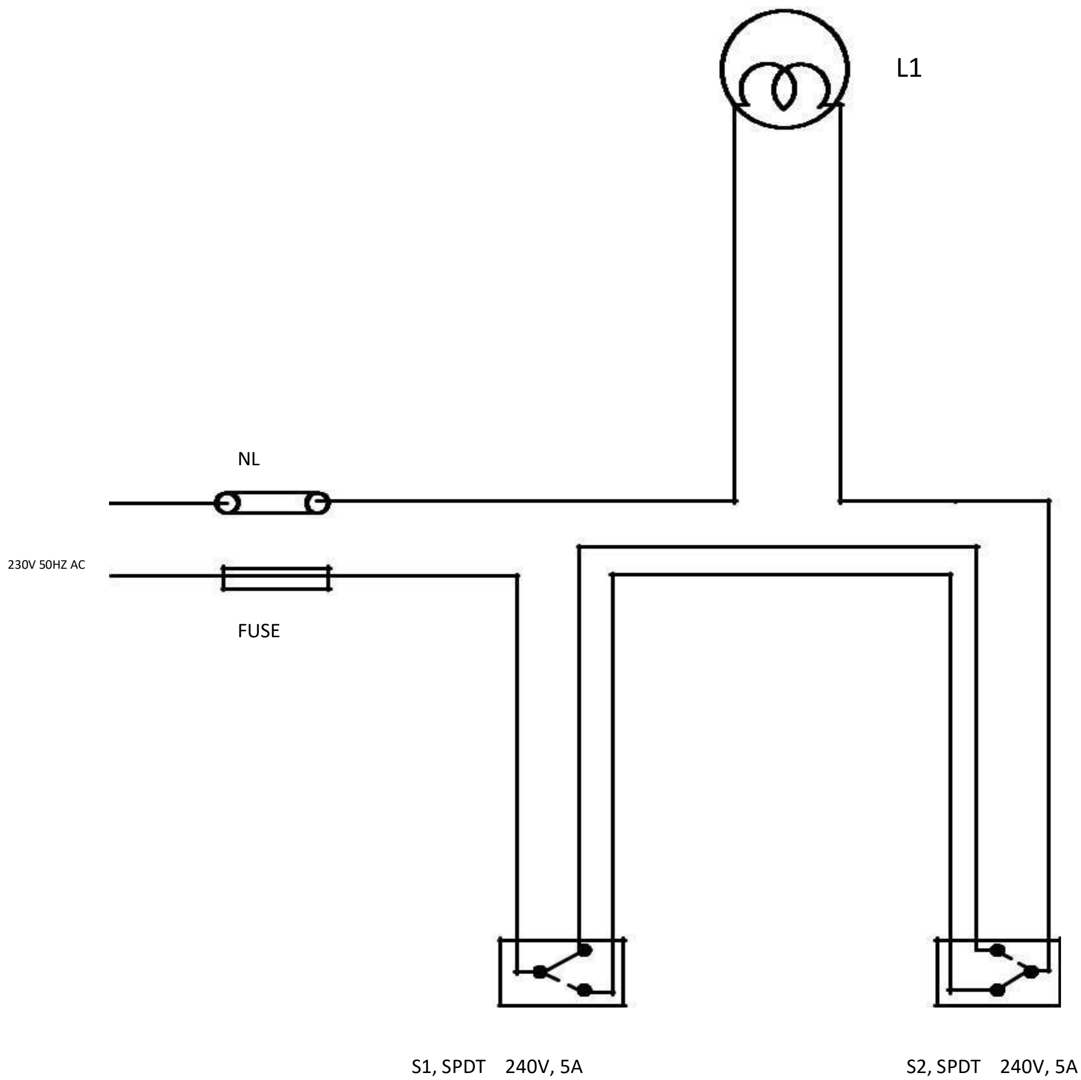
Power/Voltage =

Fuse wire rating =1.5 x Load current

LAYOUT DIAGRAM



CIRCUIT DIAGRAM



OBSERVATIONS

S1	**S2**	**LAMP**
C 1	C 1	DARK
C1	C2	BRIGHT
C2	C2	DARK
C2	C1	BRIGHT

MATERIALS REQUIRED:

SL NO	ITEMS	SPECIFICATIONS	RATING	QUANTITY
1	Conduit pipe	20mm PVC		
2	PVC insulated copper wire	$1mm^2$		
3	Fuse unit	Kitkat (ordinary type)		
4	Neutral link	Porcelain base with copper segment		
5	Saddles	20mm Metal		
6	Switch	SPST/SPDT		
7	Round block	3inch PVC		
8	Holder	Study batten		
9	20mm PVC Elbow or Junction boxes	20mm PVC(3 WAY,4 WAY)		
10	Switch boxes	3inch x 2inch PVC		
11	lamp	ordinary type incandescent		
12	Screws	½ inch ¾ inch 1 ¼ inch		

TOOLS REQUIRED:

Screw driver, working board, pocket knife, Combination pliers, pocker, Ball peen Hammer etc

RESULT:

The wiring has carried out successfully as per the given layout and conditions

HOSPITAL WIRING

AIM: Hospital wiring

Design and wire up a circuit in conduit system of wiring as per the given layout (also prepare an estimate of materials required), which satisfy the following conditions

DESIGN

Power (load) =

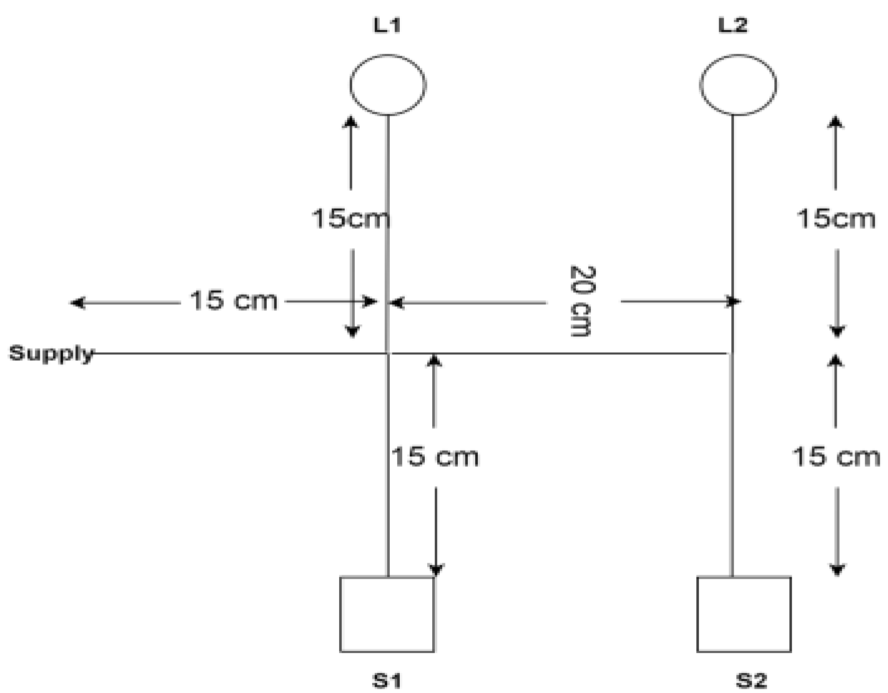
Power P =VI COSϕ

Load current

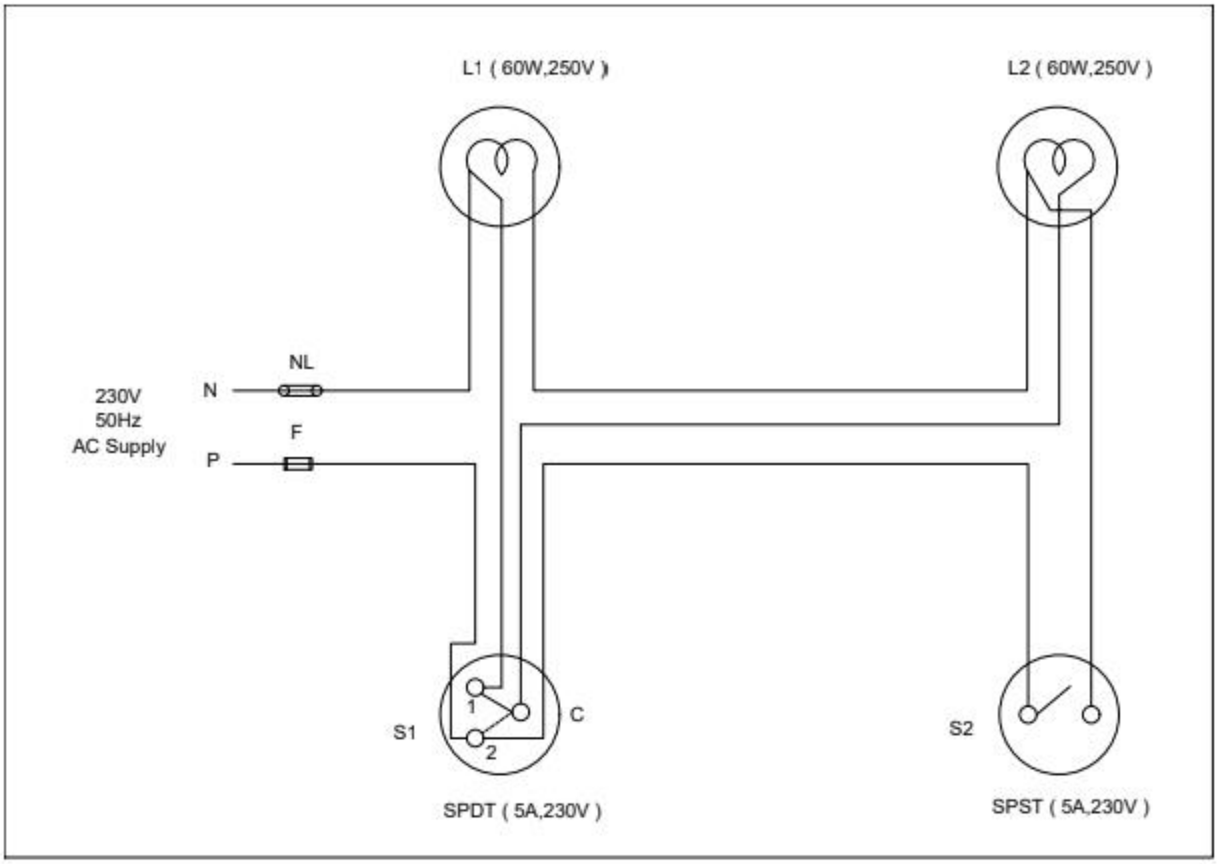
Power/Voltage =

Fuse wire rating =1.5 x Load current

LAYOUT DIAGRAM



CIRCUIT DIAGRAM



OBSERVATIONS

S1	**S2**	**LAMP**	**LAMP**
C 1	OFF	DARK	DARK
C2	OFF	DIM	DIM
C1	ON	BRIGHT	BRIGHT
C2	ON	BRIGHT	DARK

MATERIALS REQUIRED:

SL NO	ITEMS	SPECIFICATIONS	RATING	QUANTITY
1	Conduit pipe	20mm PVC		
2	PVC insulated copper wire	$1mm^2$		
3	Fuse unit	Kitkat (ordinary type)		
4	Neutral link	Porcelain base with copper segment		
5	Saddles	20mm Metal		
6	Switch	SPST/SPDT		
7	Round block	3inch PVC		
8	Holder	Study batten		
9	20mm PVC Elbow or Junction boxes	20mm PVC(3 WAY,4 WAY)		
10	Switch boxes	3inch x 2inch PVC		
11	lamp	ordinary type incandescent		
12	Screws	½ inch ¾ inch 1 ¼ inch		

TOOLS REQUIRED:

Screw driver, working board, pocket knife, Combination pliers, pocker, Ball peen Hammer etc

RESULT:

The wiring has carried out successfully as per the given layout and conditions

PLUG POINTWIRING

AIM: PLUG POINT WIRING

Design and wire up a circuit in conduit system of wiring as per the given layout (also prepare an estimate of materials required), which satisfy the following conditions

DESIGN

Power (load) = (Plug point is connected to 500W load)

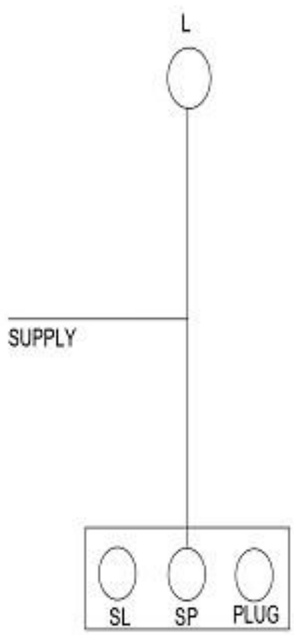
Power P =VI COSϕ

Load current

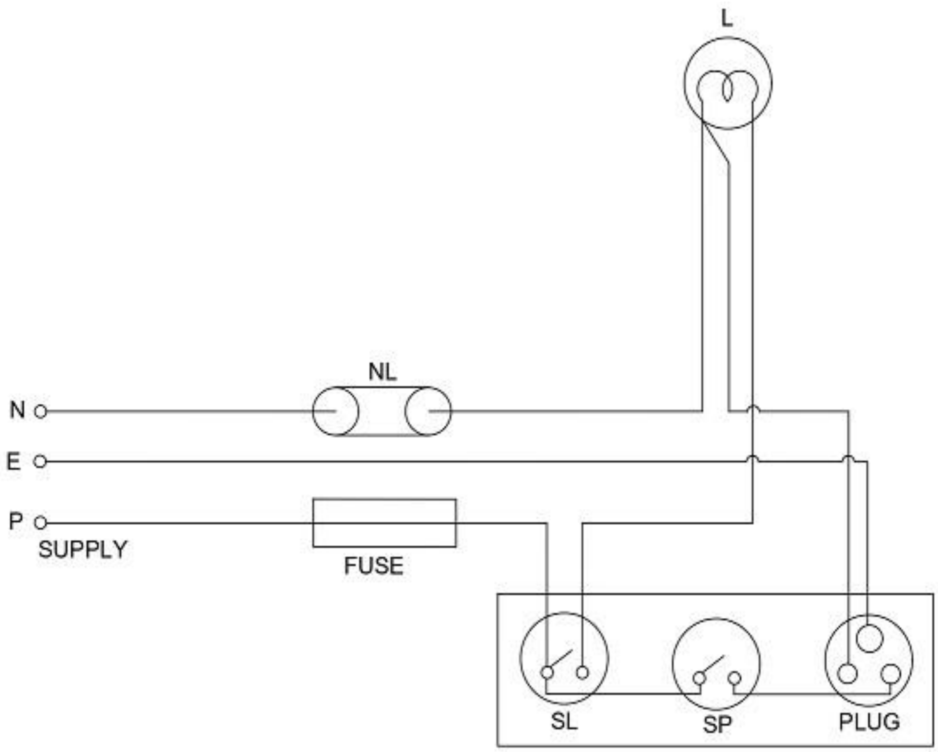
Power/Voltage =

Fuse wire rating =1.5 x Load current

LAYOUT DIAGRAM



CIRCUIT DIAGRAM



OBSERVATIONS

SP	**SL**	**PLUG**	**LAMP**
OFF	OFF	DEAD	DARK
ON	OFF	LIVE	DARK
OFF	ON	DEAD	BRIGHT
ON	ON	LIVE	BRIGHT

MATERIALS REQUIRED:

SL NO	ITEMS	SPECIFICATIONS	RATING	QUANTITY
1	Conduit pipe	20mm PVC		
2	PVC insulated copper wire	$1mm^2$		
3	Fuse unit	Kitkat (ordinary type)		
4	Neutral link	Porcelain base with copper segment		
5	Saddles	20mm Metal		
6	Switch	SPST/SPDT		
7	Round block	3inch PVC		
8	Holder	Study batten		
9	20mm PVC Elbow or Junction boxes	20mm PVC(3 WAY,4 WAY)		
10	Switch boxes	6inch x 4inch PVC		
11	lamp	ordinary type incandescent		
12	Screws	½ inch ¾ inch 1 ¼ inch		

TOOLS REQUIRED:

Screw driver, working board, pocket knife, Combination pliers, pocker, Ball peen Hammer etc

RESULT:

The wiring has carried out successfully as per the given layout and conditions

Single phase distribution board

AIM: Single phase distribution board

Design and wire up a circuit in conduit system of wiring as per the given layout (also prepare an estimate of materials required), which satisfy the following conditions

DESIGN

Power (load) =

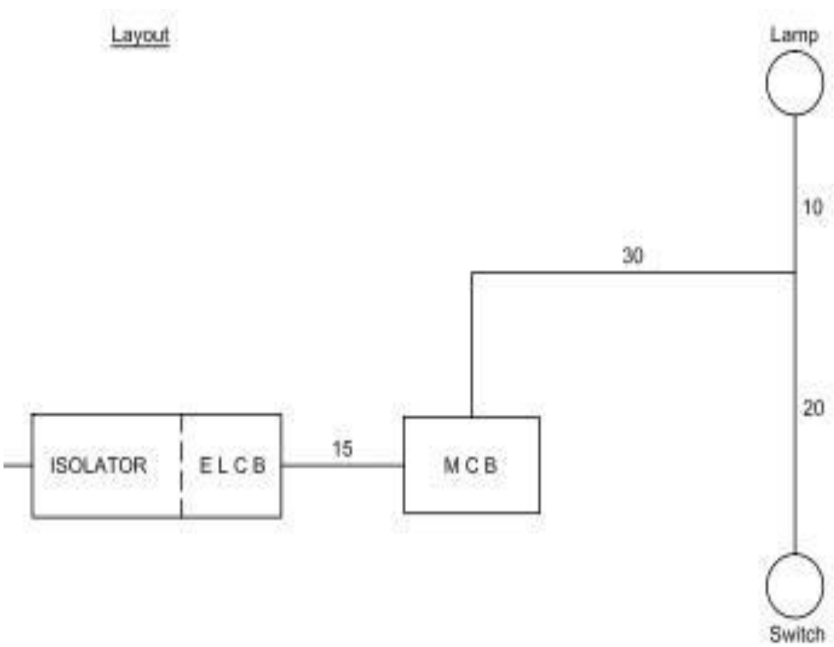
Power P =VI COSϕ

Load current

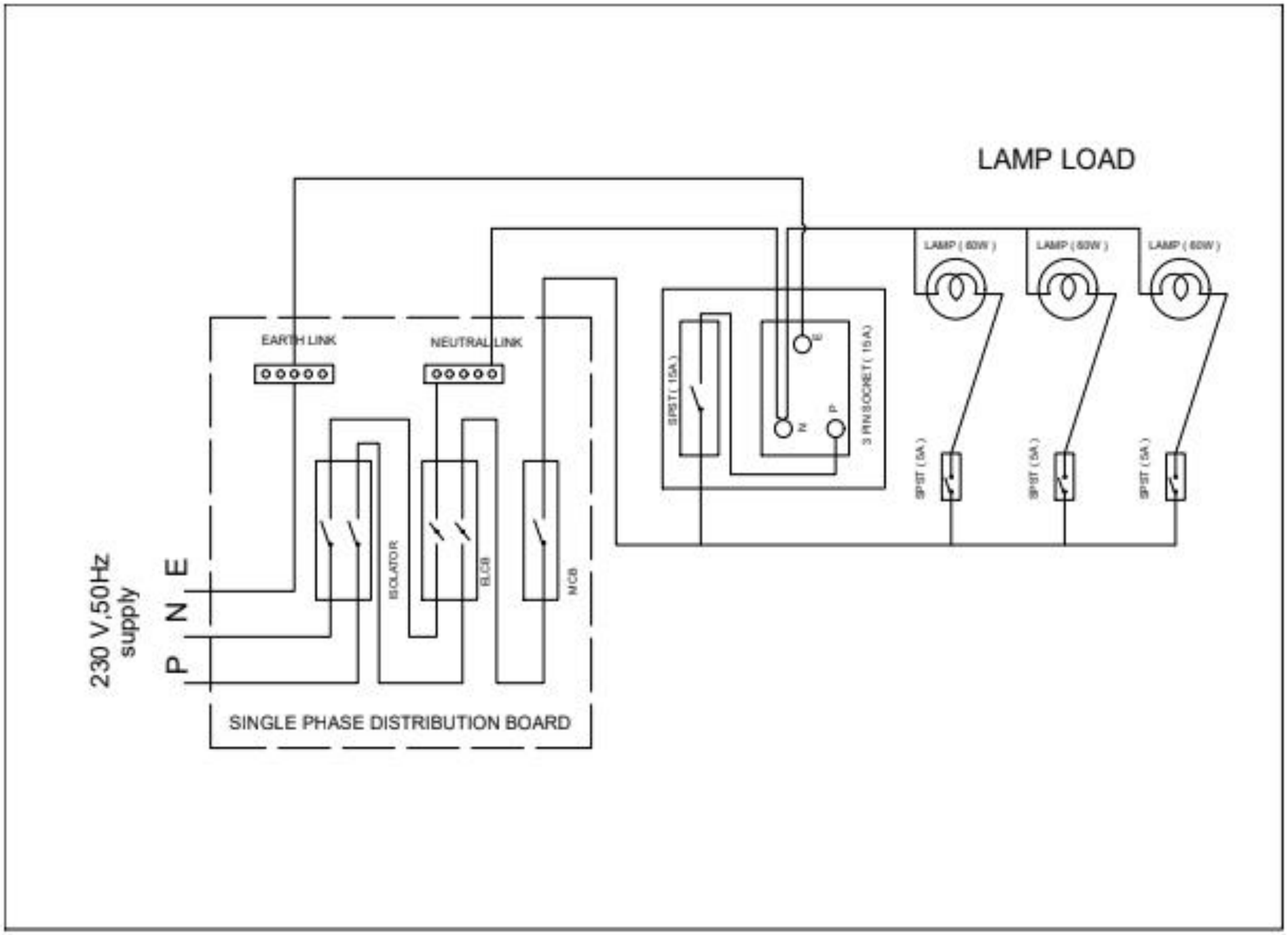
Power/Voltage =

Fuse wire rating =1.5 x Load current

LAYOUT DIAGRAM



CIRCUIT DIAGRAM



LAMP LOAD
LAMP (60W)
LAMP (60W)
LAMP (60W)
SPST (5A)
SPST (5A)
SPST (5A)
SPST (15A)
3 PIN SOCKET (15A)
E
N
P
EARTH LINK
NEUTRAL LINK
ISOLATOR
ELCB
MCB
SINGLE PHASE DISTRIBUTION BOARD
230 V,50Hz
supply
P
N
E

MATERIALS REQUIRED:

SL NO	ITEMS	SPECIFICATIONS	RATING	QUANTITY
1	Conduit pipe	20mm PVC		
2	PVC insulated copper wire	1mm^2	5A	
3	Fuse unit	Kitkat (ordinary type)	16 A,250V	
4	Neutral link	Porcelain base with copper segment	16 A,250V	
5	Saddles	20mm Metal		
6	Switch	SPST/SPDT	5A	
7	ELCB		40 A in 30 ma	
8	MCB		10A	
9	Isolator		40A	
10	Switch boxes	3inch x 2inch PVC		
11	lamp	ordinary type incandescent	60W	
12	Screws	½ inch ¾ inch 1 ¼ inch		As per the given layout

TOOLS REQUIRED:

Screw driver, working board, pocketknife, pliers, pocker, hacksaw, connector, chisel, mallet, steel rule, etc

RESULT:

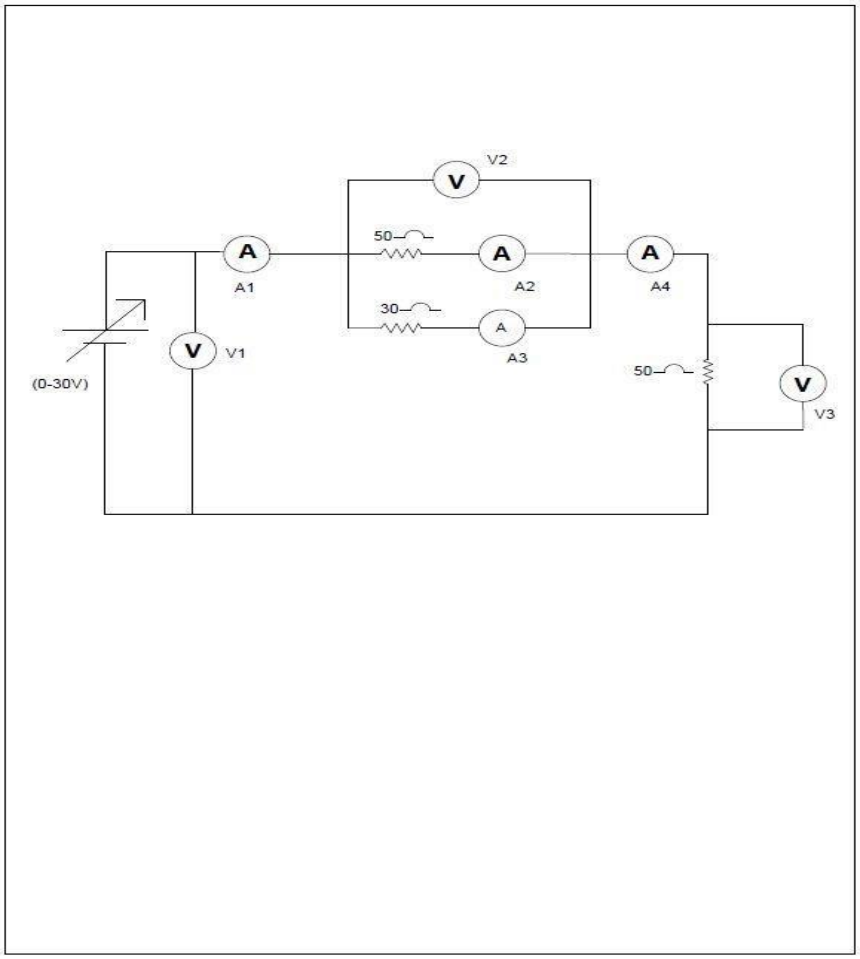
The wiring has carried out successfully as per the given layout and conditions

VERIFICATION OF BASIC CIRCUIT LAWS

Aim:

To experimentally verify Ohm's Law & Kirchoff's Laws for DC circuits.

CIRCUIT DIAGRAM



Procedure

1. Wire up the circuit as shown in the diagram.
2. Record the readings of the meters for various values of the input voltage.
3. Based upon the meter readings, computations are performed for verifying Ohm's Law & Kirchoff's Laws

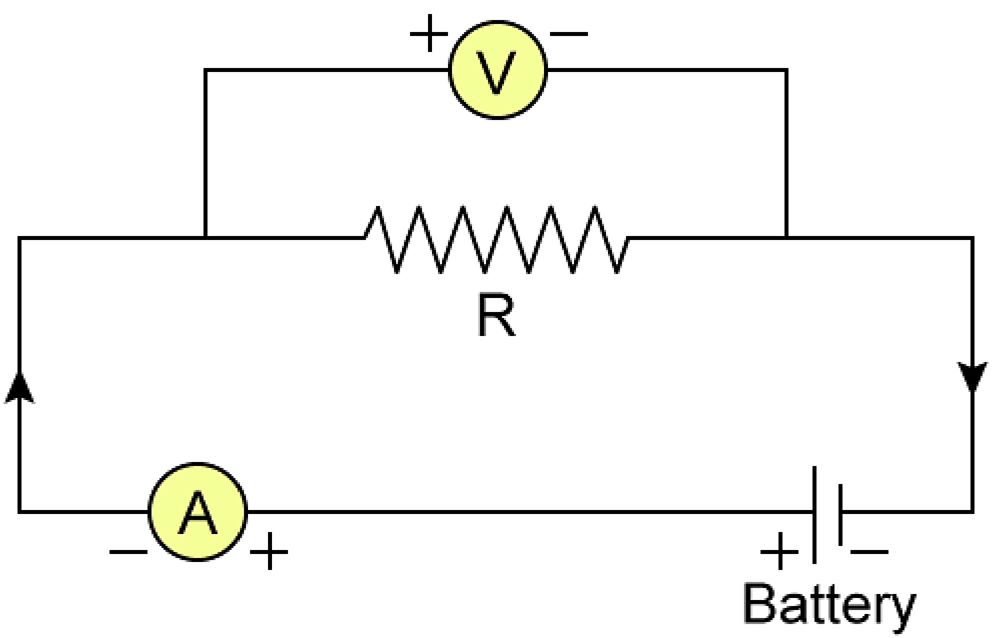
Tabulation.

V1	V2	V3	A1	A2	A3	A4	V2+V3	A2+A3

Sample Calculation ***(Set No: ....)***

(Verify that V1=V2+V3 & A1=A2+A3)

CIRCUIT DIAGRAM



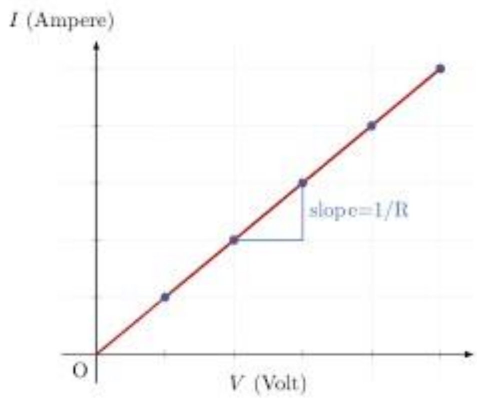
Tabulation.

SL NO	VOLTAGE (V)	CURRENT (A)

Sample Calculation

V = IR

Sample graph (V&I)



Result

The experiment has been completed successfully.

(Theoretical Background required for completing the Experiment: Ohm’s Law, KCL, KVL)

Study of Cathode Ray Oscilloscope

Objective: To understand the operation of the CRO and to learn how to determine the Amplitude Time period and Frequency of a given waveform using CRO

Apparatus:

S.No	Apparatus	Type	Range	Quantity
01	CRO			01
02	Function Generator		10-1MHz	01
03	Regulated Power supply		(0-30V)	01
04	Audio frequency probe			01

Introduction: CRO is an electronic device which is capable of giving a visual indication of a signal waveform. With an oscilloscope the waveform of the signal can be studied with respect to amplitude distortion and deviation from the normal. Oscilloscope can also be used for measuring voltage, frequency and phase shift.

Cathode Ray Tube**:** Cathode Ray Tube is a heart of Oscilloscope providing visual display of the input signals. CRT consists of three basic parts.

1. Electron Gun.

2.Deflecting System.

3.Flouroscent

Screen These essential parts are arranged inside a

tunnel shaped glass envelope.

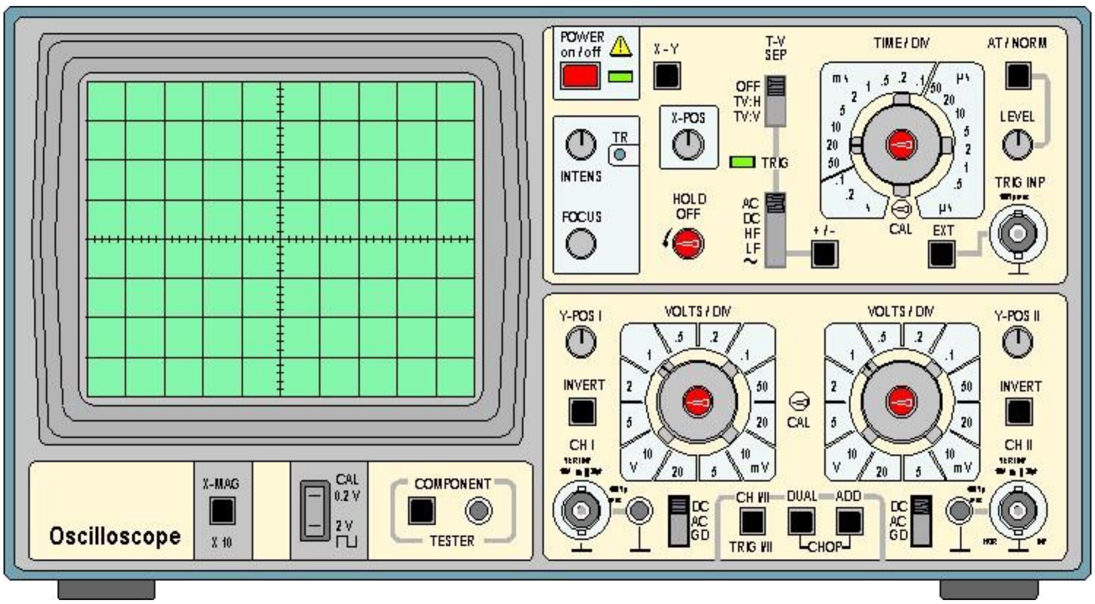
Electron Gun**:** The function of this is to provide a sharply focused stream of electrons. It mainly consists of an indirectly heated cathode, a control grid, focusing anode and accelerating anode. Control grid is cylinder in shape. It is connected to negative voltage w.r.t to cathode. Focusing and accelerating anodes are at high positive potential. w.r.t anode. Cathode is indirectly heated type & is heated by filament. Plenty of electrons are released from the surface of cathode due to Barium Oxide coating. Control Grid encloses the cathode and controls the number of electrons passing through the tube.

A voltage on the control grid consists the cathode determines the number of electrons freed by heating which are allowed to continue moving towards the face of the tube. The accelerated anode is heated at much higher potential than focusing anode. Because of this reason the accelerating anode accelerates the light beam into high velocity. The beam when strikes the screen produces the spot or visible light.

The name electron Gun is used because it fires the electrons like a gun that fires a bullet.

Deflection system**:** The beam after coming out of the accelerated anode passes through two sets of deflection plates with the tube . The first set is the vertical deflection plate and the second set is horizontal deflection plates. The vertical deflection plates are oriented to deflect the electron beam that moves vertically up and down. The direction of the vertical deflection beam is determined by the voltage polarity applied to the plates. The amount of deflection is set by the magnitude of the applied voltage. The beam is also deflected horizontally left or right by a voltage applied to horizontal plates. The deflecting beam is then further accelerated by a very high voltage applied to the tube.

Fluorescent Screen**:** The screen is large inside the face of the tube and is coated with a thin layer of florescent material called Phosphor. On this fluorescent material when high velocity electron beam strikes its converting the energy of the electron the electron beam between into visible light(spots). Hence the name is given as fluorescent screen.



PANEL CONTROLS:

1. POWER ON/OFF : Push the button switch to supply power to the instrument.

2. X5 : Switch when pushed inwards gives 5 times magnification of the X signal

3. XY : Switched when pressed cut off the time base and allows access the exit horizontal signal to be fed through CH II

(used for XY display).

4. CH I/CH II/TRIG I/ TRIG II. : Switch out when selects and triggers CH I and when Pressed selects and triggers CH II.

5. MOD/DUAL : Switch when selects the dual operation switch

6. ALT/CHOP/ADD : Switch selects alternate or chopped in dual mode. If mode is selected then this switch enables addition or subtraction of the channel i.e. CH-! +- CH II.

7. TIME/DIV	**:** Switch selects the time base speed.
8. AT/NORM	**:** Switch selects AUTO/NORMAL position .Auto is used to get trace when no signal is fed at the input . In NORM the trigger level can be varied from the positive peak to negative peak with level control.
9. LEVEL	**:** Controls the trigger level from the peak to peak amplitude signal.
10. TRIG.INP	**:** Socket provided to feed the external trigger signal in EXT. mode.
11. CAL OUT	**:** Socket provided for the square wave output 200 mv used for probe compensation and checking vertical sensitivity etc.
12. EXT	**:** Switch when pressed allows external triggering signal to be fed from the socket marked TRIG.INP.
13. X-POS	**:** Controls the horizontal position of the trace.
14. VAR	**:** Controls the time speed in between two steps of time/div switch .For calibration put this fully anticlockwise (at cal pos)
15. TV	**:** Switch when it allows video frequency up to 20 KHz to be locked.
16. + -	: Switch selects the slope of trigger whether positive going or negative.
17. INV CHJ II	**:** Switch when pressed inverts the CH ii.
18. INTENS	**:** Controls brightness of trace.
19. TR	**:** Controls the alignment of the trace with gratitude
20. FOCUS	**:** Controls the sharpness of the trace.
21. CT	: Switch when pressed starts CT operation.
22. GD/AC /DC	: Input coupling switch for each channel. In AC the signal is coupled through the 0.1 MFD capacitor.
23. DC/AC/GD	**:** BNC connectors serve as input connectors for the CH I and CH II channel input connector also serves as the horizontal external signal.
24. CT-IN	**:**To test any components in the circuit, put one test probe in this socket and connect the other test probe in the ground socket.
25. VOLTS /DIV	: Switches select the sensitivity of each channel.
26. Y POS I AND II	**:** Controls provided for vertical deflection for each channel.

BACK PANEL CONTROLS

1. FUSE	: 350 mA fuse is provided at the back panel spare fuses are provided

inside the instrument.

2.ZMOD **:** Banana socket provided for modulating signal input i.e. Z-modulation.

Precautions

1. Avoid using CRO in high ambient light conditions.

2. Select the location free from Temperature & humidity. It should not be used in dusty environment.

3. Do not operate in a place where mechanical vibrations are more or in a place which generates strong magnetic fields or impulses.

5. Do not increase the brightness of the CRO than that is required.

Experiment:

1. Turn on the power of the CRO.

2. From the Function Generator select the desired frequency and amplitude of the sine wave.

3. The amplitude of the waveform is obtained by noting the number of divisions along the Y-axis in between peak to peak of the waveform (i.e. sine waveform / Triangular waveform /Square waveform) and multiplying with the divisional factor of the amplitude note in volts.

4. Time period is calculated from X-axis.

5. Frequency is obtained by formula F=1/T.
6. This frequency is compared with the frequency applied using function generator.

7. Voltage in the CRO is compared with the voltage applied from function generator.

8. By repeating the above steps we can find frequency and voltages of square wave & triangular waveforms.

Tabular Column:

Waveform	Time Period(sec)		Frequency(Hz)		Amplitude(V)	
	Theoretical	Practical	Theoretical	Practical	Theoretical	Practical
Sinusoidal						
Triangular						
Square						

Calculations:

1. Sinusoidal Waveform:

Amplitude: __V

Time Period: ___Sec

Frequency: Hz

2. Square Waveform:

Amplitude: __V

Time Period: ___Sec

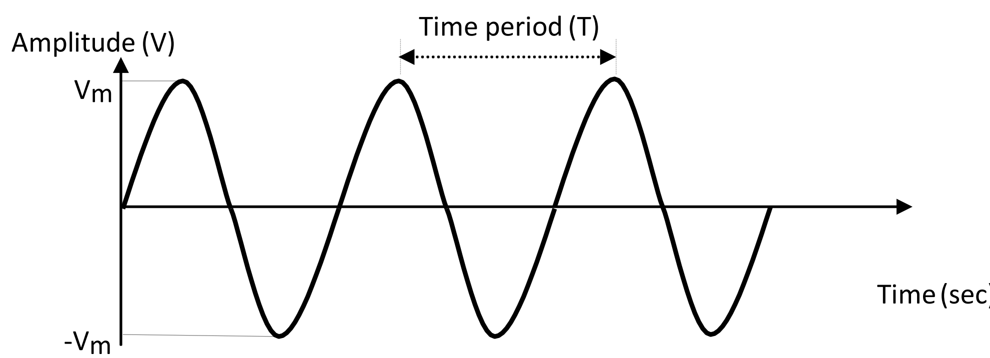
Frequency: Hz

3. Triangular Waveform:

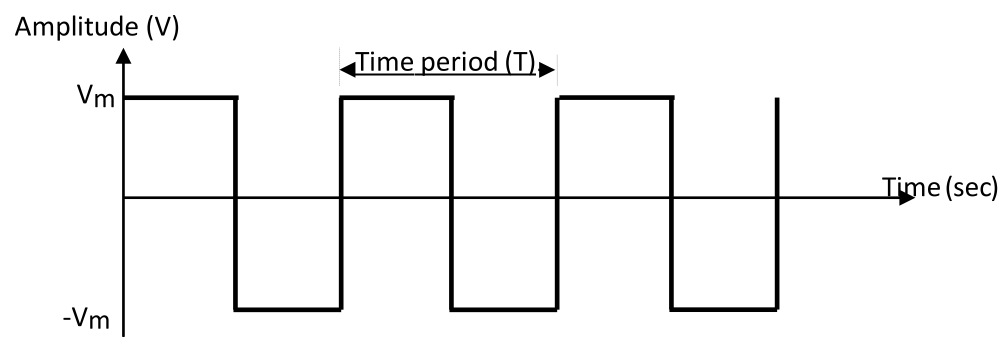
Amplitude: _____V
Time Period:_____Sec

Frequency: ____Hz

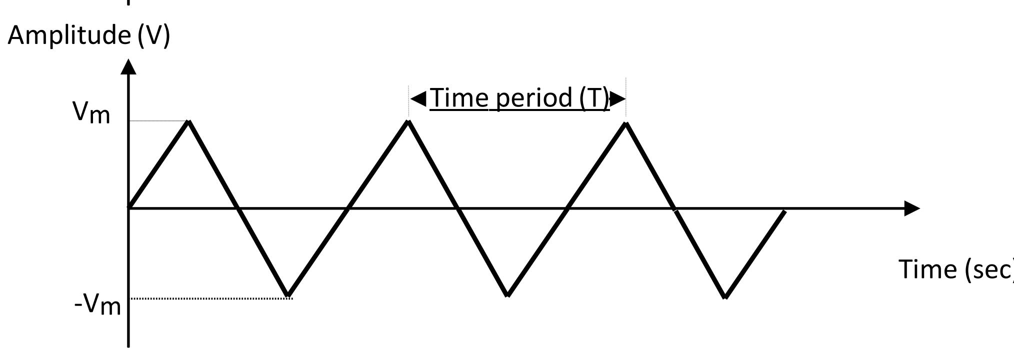
MODEL GRAPHS:



Sine wave



Square wave



Triangular wave

Half-Wave rectifier with and without filter

Objective

1. To plot input and output waveforms of the Half Wave Rectifier with and without Filter
2. To find ripple factor of Half Wave Rectifier with and without Filter
3. To find percentage regulation of Half Wave Rectifier with and without Filter

Apparatus

S.No	Apparatus	Type	Range	Quantity
01	Transformer	Step-down	0-12V	01
02	Diode	IN4007		01
03	Decade Resistance Box		10-1KΩ	01
04	Capacitor		1000µF/25V	01
05	Digital Multimeter(DMM)		(0-20V)	01
06	CRO & CRO Probes			01
07	Breadboard and Wires			

INTRODUCTION:

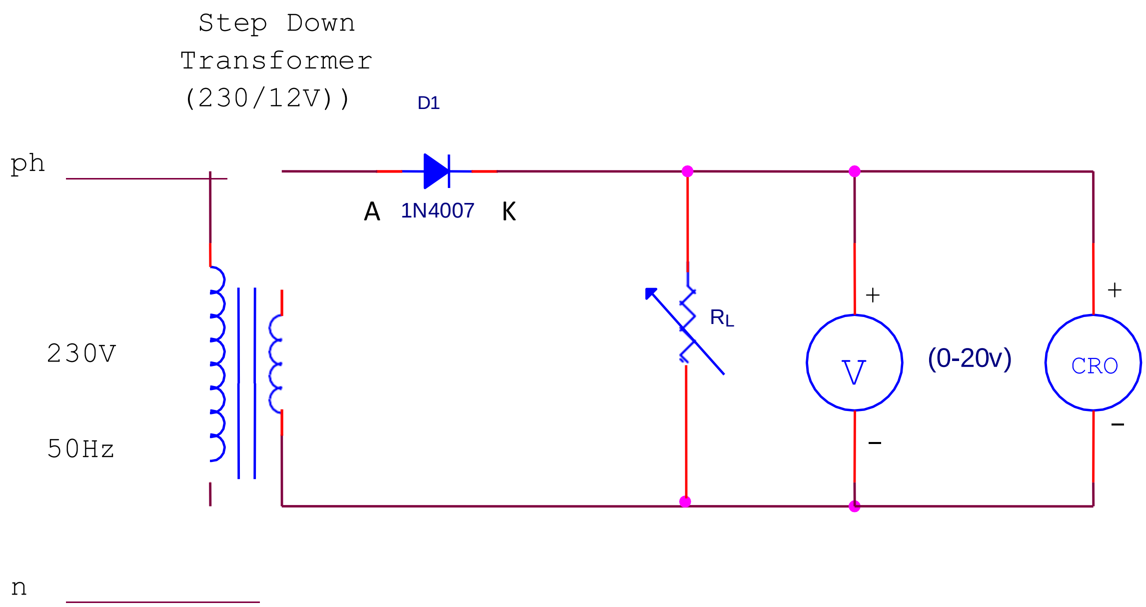
A device is capable of converting a sinusoidal input waveform into a unidirectional waveform with non zero average component is called a rectifier.

A practical half wave rectifier with a resistive load is shown in the circuit diagram. During the positive half cycle of the input the diode conducts and all the input voltage is dropped across R_L. During the negative half cycle the diode is reverse biased and it acts as almost open circuit so the output voltage is zero.

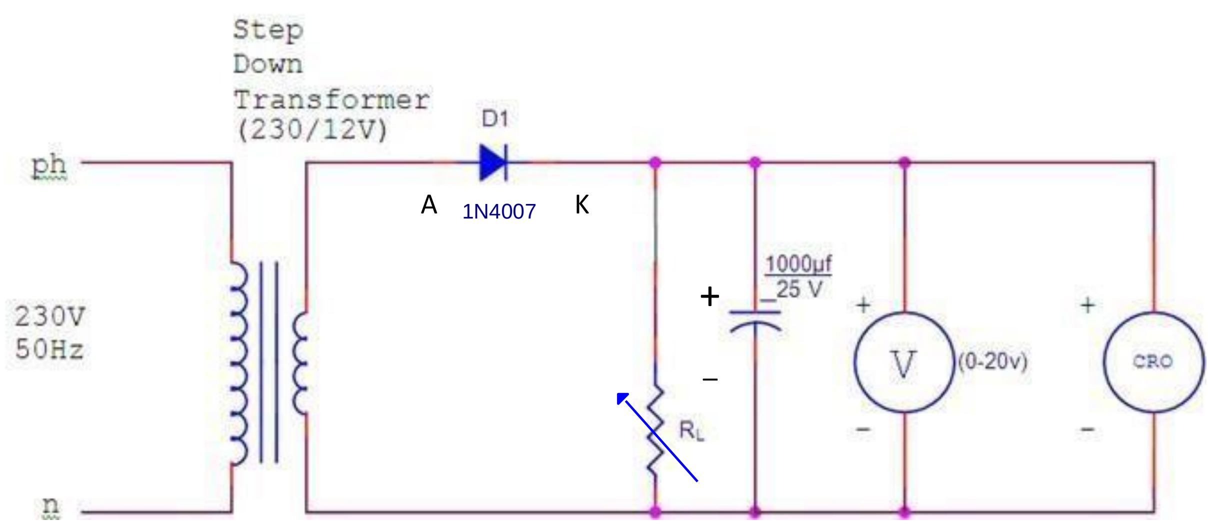
The filter is simply a capacitor connected from the rectifier output to ground. The capacitor quickly charges at the beginning of a cycle and slowly discharges through R_L after the positive peak of the input voltage. The variation in the capacitor voltage due to charging and discharging is called ripple voltage. Generally, ripple is undesirable, thus the smaller the ripple, the better the filtering action.

Circuit Diagram

Without Filter



With Filter



PRECAUTIONS

1. The primary and secondary sides of the transformer should be carefully identified.
2. The polarities of the diode should be carefully identified.

Theoretical calculations for Ripple factor:-

Without Filter:-

$$V_{dc} = \frac{V_m}{\pi}$$

$$V_{rms} = \frac{V_m}{2}$$

$$\text{Ripple factor} = \sqrt{\left(\frac{V_{rms}}{V_{dc}}\right)^2 - 1} = 1.21$$

With Filter:-

$$\text{Ripple factor} = \frac{1}{2\sqrt{3}\, fCR_L} =$$

Where f =50Hz
C =1000µF
R_L=1KΩ

EXPERIMENT (without Filter)

1. Connections are made as per the circuit diagram of the rectifier without filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3.Note down the no load voltage before applying the load to the Circuit and by using the Multimeter, measure the ac input voltage of the rectifier and its frequency.

4.Now Vary the R_L in steps of 100Ω by varying the DRB from 1100Ω to 100Ω and note down the load voltage (V_L) using the multimeter for each value of R_L and calculate the percentage regulation.

5.Measure the AC and DC voltage at the output of the rectifier for each value of R_L using Multimeter.

6.Now Observe the output waveform on CRO across R_L and find out value of V_m.

7. Now calculate V_{dc}, V_{rms}, Ripple Factor and other parameters of half wave rectifier according to thegiven formulae.

8.Measure the amplitude and timeperiod of the transformer secondary(input waveform) by connectingCRO.

9.Feed the rectified output voltage to the CRO and measure the time period and amplitude of thewaveform.

EXPERIMENT (with Filter)

1. Connections are made as per the circuit diagram of the rectifier with filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output ofthe rectifier.

4.Measure the amplitude and timeperiod of the transformer secondary(input waveform) by connectingCRO.

5.Feed the rectified output voltage to the CRO and measure the time period and amplitude of thewaveform.

Tabular Column: **Without Filter**

Using DMM:

V_{ac}	V_{dc}	**Ripple Factor(ɤ)=** V_{ac}/V_{dc}

Using CRO :

R_L (Ω)	V_L (V)	V_m (V)	$Vdc = \frac{Vm}{\pi}$ (V)	$Vrms = \frac{Vm}{2}$ (V)	$Vr(rms) = \sqrt{Vrms^2 - Vdc^2}$ (V)	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation = $\frac{V(NL)-V(L)}{VL}$

With Filter

Using DMM:

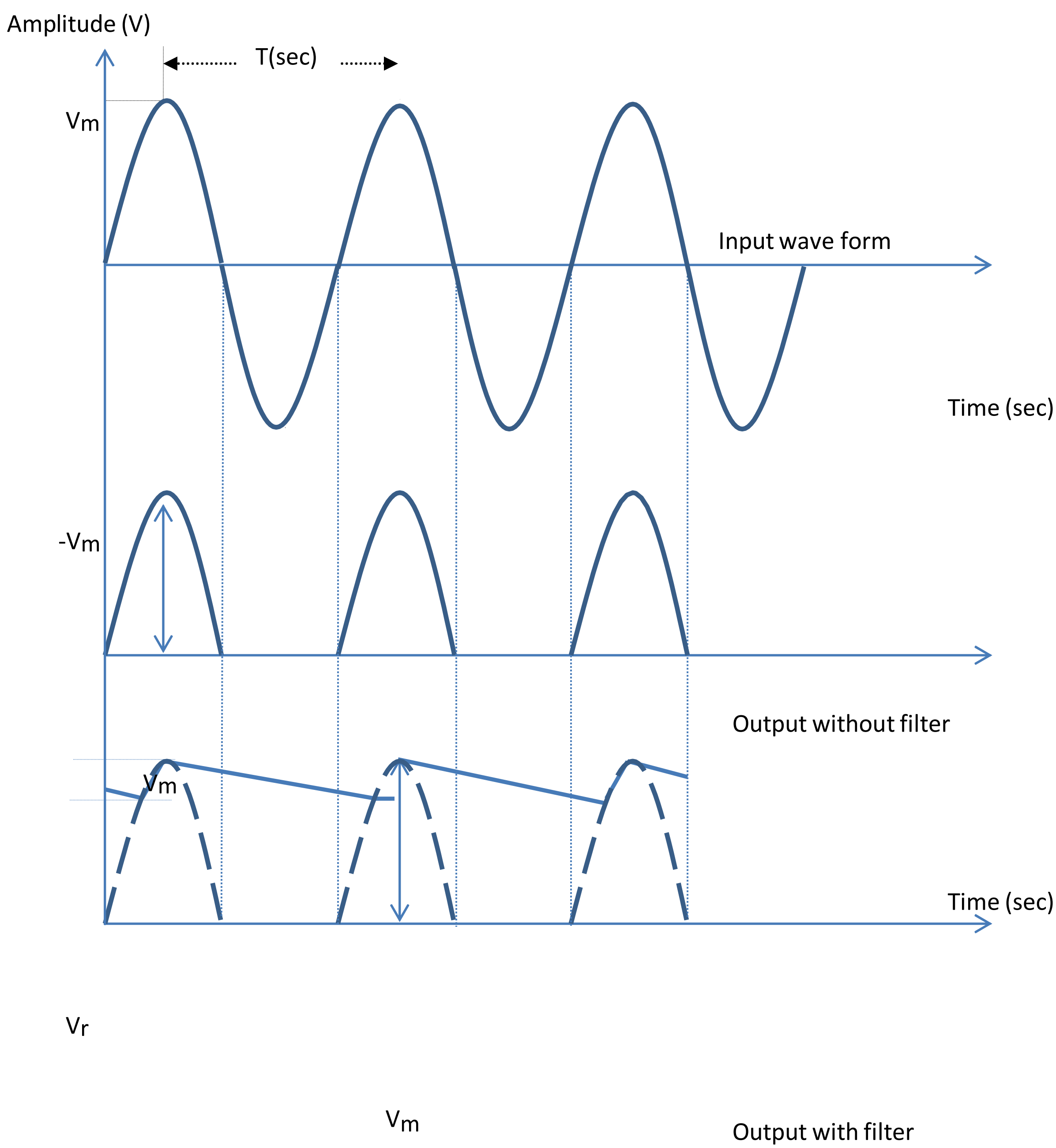
V_{ac}	V_{dc}	**Ripple Factor(ɤ)=** V_{ac}/V_{dc}

Using CRO :

V_{NL}=

R_L (Ω)	V_L (V)	V_m (V)	V_r (V)	$Vdc = Vm - \frac{Vr}{2}$ (V)	$Vr(rms) = \frac{Vr}{2\sqrt{3}}$	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation = $\frac{V(NL)-V(L)}{VL}$

OUTPUT WAVEFORMS:



Result: The input and output waveforms of half wave rectifier is plotted and the ripple factor and regulation at 1100Ω are

Ripple factor with out Filter =

Ripple factor with Filter =

%Regulation=

Full-Wave rectifier with and without filter

Objective

1. To plot input and output waveforms of the Full Wave Rectifier with and without Filter
2. To find ripple factor for Full Wave Rectifier with and without Filter
3. To find regulation for Full Wave Rectifier with and without Filter

Apparatus

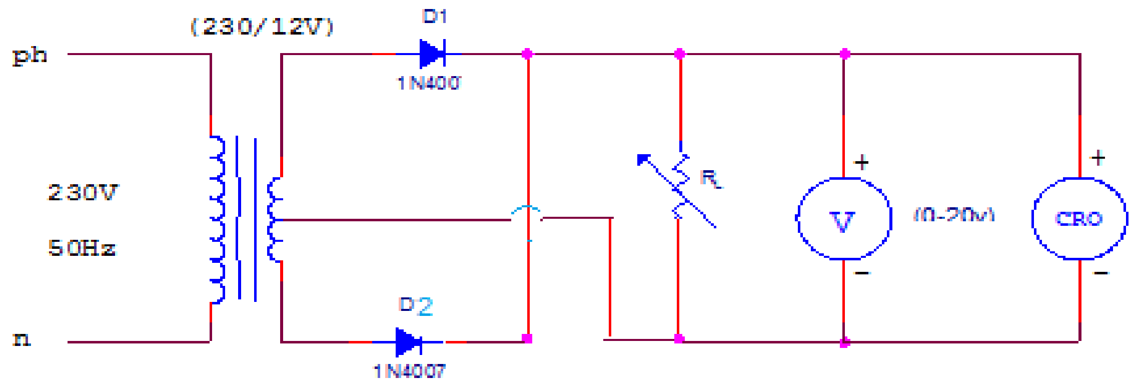
S.No	Apparatus	Type	Range	Quantity
01	Transformer	Centertapped	12-0-12V	01
02	Diode	IN4007		02
03	Resistance		1KΩ	01
04	Capacitor		1000µF/25V	01
05	Multimeter		(0-20V)	01
06	CRO			01
07	Breadboard and Wires			

INTRODUCTION:

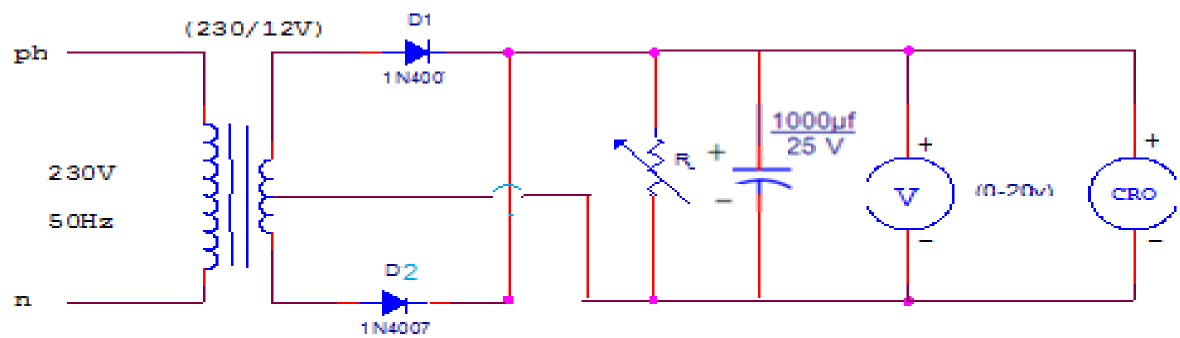
A device is capable of converting a sinusoidal input waveform into a unidirectional waveform with non zero average component is called a rectifier. A practical half wave rectifier with a resistive load is shown in the circuit diagram. It consists of two half wave rectifiers connected to a common load. One rectifies during positive half cycle of the input and the other rectifying the negative half cycle. The transformer supplies the two diodes (D1 and D2) with sinusoidal input voltages that are equal in magnitude but opposite in phase. During input positive half cycle, diode D1 is ON and diode D2 is OFF. During negative half cycle D1 is OFF and diode D2 is ON. Generally, ripple is undesirable, thus the smaller the ripple, the better the filtering action.

Circuit Diagram

Without Filter



With Filter



PRECAUTIONS

1. The primary and secondary sides of the transformer should be carefully identified.
2. The polarities of the diode should be carefully identified.

<u>Theoretical calculations for Ripple factor:-</u>

<u>Without Filter:-</u>

$$V_{dc} = \frac{2V_m}{\pi}$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$\text{Ripple factor} = \sqrt{\left(\frac{V_{rms}}{}\right)^2 - 1} \quad =0.482$$

<u>With Filter</u>:-

$$\text{Ripple factor} = \frac{1}{4\sqrt{3}fCR_L} =$$

Where f =50Hz

C =1000µF

R_L=1KΩ

Experiment(without filter)

1. Connections are made as per the circuit diagram of the rectifier without filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output ofthe rectifier.

4. Measure the amplitude and time period of the transformer secondary (input waveform) by connectingCRO.

5. Feed the rectified output voltage to the CRO and measure the time period and amplitude of theWaveform.

Experiment (With filter)

1. Connections are made as per the circuit diagram of the rectifier with filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output ofthe rectifier.

4. Measure the amplitude and time period of the transformer secondary (input waveform) by connectingCRO.

5. Feed the rectified output voltage to the CRO and measure the time period and amplitude of thewaveform.

Tabular Column Without Filter

Using DMM:

V_{ac}	V_{dc}	Ripple Factor(γ)= V_{ac}/ V_{dc}

Using CRO :

V_{NL}=

R_L (Ω)	V_L (V)	V_m (V)	$Vdc = \frac{2Vm}{\pi}$ (V)	$Vrms = \frac{Vm}{\sqrt{2}}$ (V)	$Vr(rms) = \sqrt{Vrms^2 - Vdc^2}$ (V)	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation $= \frac{V(NL)-V(L)}{VL}$

With Filter

Using DMM:

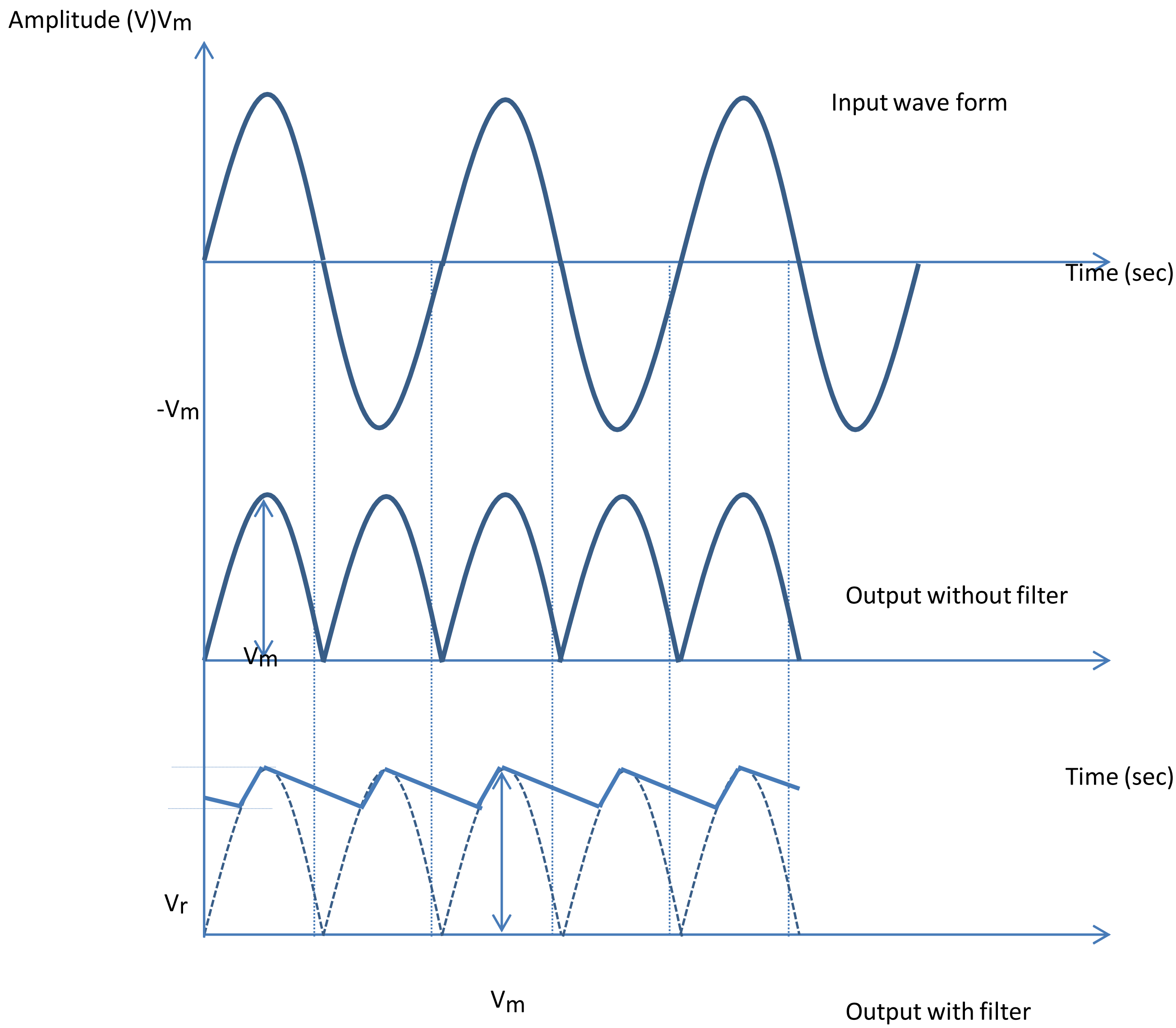
V_{ac}	V_{dc}	Ripple Factor(ɤ)= V_{ac}/ V_{dc}

Using CRO :

V_{NL}=

R_L (Ω)	V_L (V)	V_m (V)	V_r (V)	$Vdc = Vm - \frac{Vr}{2}$ (V)	$Vr(rms) = \frac{Vr}{4\sqrt{3}}$	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation $= \frac{V(NL)-V(L)}{VL}$

Model Graph



Result: The input and output waveforms of Full wave rectifier is plotted and the ripple factor andregulation at 1100Ω are

Ripple factor with out Filter =

Ripple factor with Filter =

%Regulation=

Bridge rectifier with and without filter

Objective

1. To plot input and output waveforms of the Bridge Rectifier with and without Filter
2. To find ripple factor for Bridge Rectifier with and without Filter
3. To find Regulation factor for Bridge Rectifier with and without Filter

Apparatus

S.No	Apparatus	Type	Range	Quantity
01	Transformer	Step-down	12-0-12V	01
02	Diode	IN4007		04
03	Resistance		1KΩ	01
04	Capacitor		1000µF/25V	01
05	Voltmeter		(0-20V)	01
06	CRO			01
07	Breadboard and Wires			

INTRODUCTION:

A device is capable of converting a sinusoidal input waveform into a unidirectional waveform with non zero average component is called a rectifier.

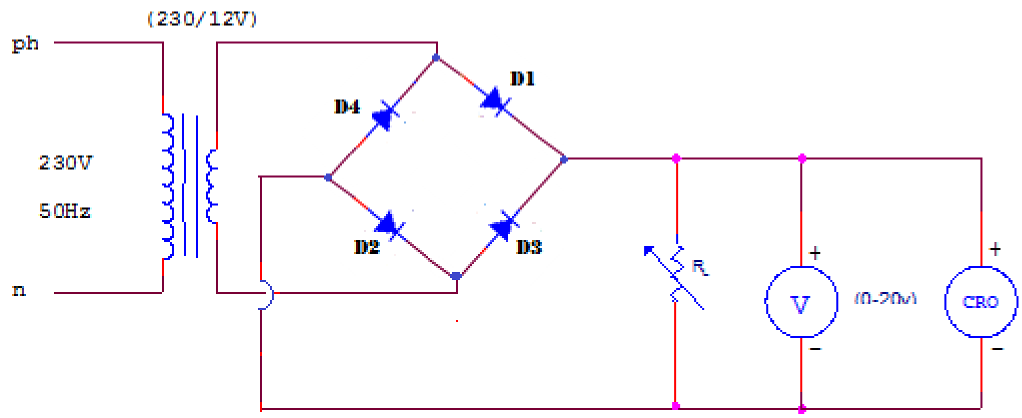
The Bridge rectifier is a circuit, which converts an ac voltage to dc voltage using both half cycles of the input ac voltage. The Bridge rectifier has four diodes connected to form a Bridge. The load resistance is connected between the other two ends of the bridge.

For the positive half cycle of the input ac voltage, diode D1 and D3 conducts whereas diodes D2 and D4 remain in the OFF state. The conducting diodes will be in series with the load resistance RL and hence the load current flows through R_L .

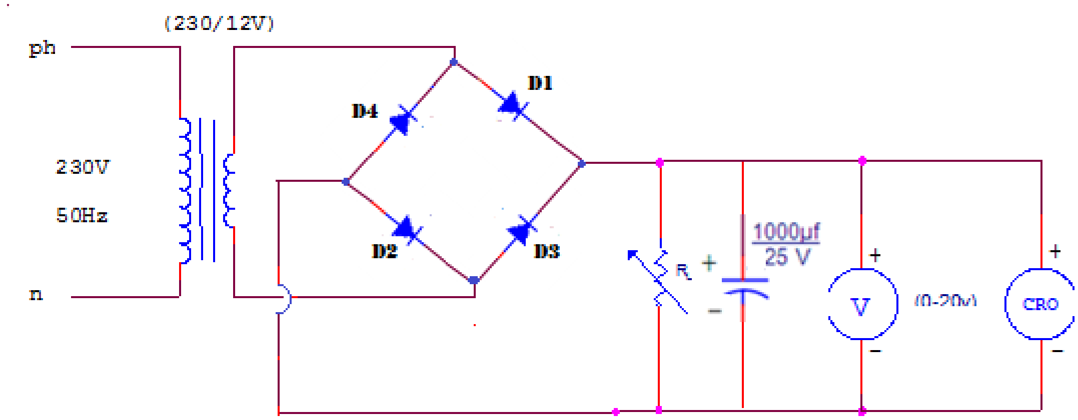
For the negative half cycle of the input ac voltage, diode D2 and D4 conducts whereas diodes D1 and D3 remain in the OFF state. The conducting diodes will be in series with the load resistance RL and hence the load current flows through RL in the same direction as in the previous half cycle. Thus a bidirectionalwave is converted into a unidirectional wave.

Circuit Diagram

Without Filter



With Filter



<u>Theoretical calculations for Ripple factor:-</u>

<u>Without Filter:-</u>

$$V_{dc} = \frac{2V_m}{\pi}$$

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

$$\text{Ripple factor } = \sqrt{\left(V_{rms}\right)^2 - 1} \quad = 0.482$$

<u>With Filter</u>:-

$$\text{Ripple factor } = \frac{1}{4\sqrt{3}\, fCR_L} =$$

Where f = 50Hz
C = 1000µF
R_L = 1KΩ

Experiment (without filter)

1. Connections are made as per the circuit diagram of the rectifier without filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output ofthe rectifier.

4. Measure the amplitude and timeperiod of the transformer secondary(input waveform) by connectingCRO.

5. Feed the rectified output voltage to the CRO and measure the time period and amplitude of thewaveform.

Experiment (With filter)

1. Connections are made as per the circuit diagram of the rectifier with filter.

2. Connect the primary side of the transformer to ac mains and the secondary side to the rectifier input.

3. By the multimeter, measure the ac input voltage of the rectifier and, ac and dc voltage at the output ofthe rectifier.

4. Measure the amplitude and timeperiod of the transformer secondary(input waveform) by connectingCRO.

5. Feed the rectified output voltage to the CRO and measure the time period and amplitude of thewaveform.

Tabular Column: **Without Filter**

Using DMM:

V_{ac}	V_{dc}	Ripple Factor(γ)= V_{ac}/ V_{dc}

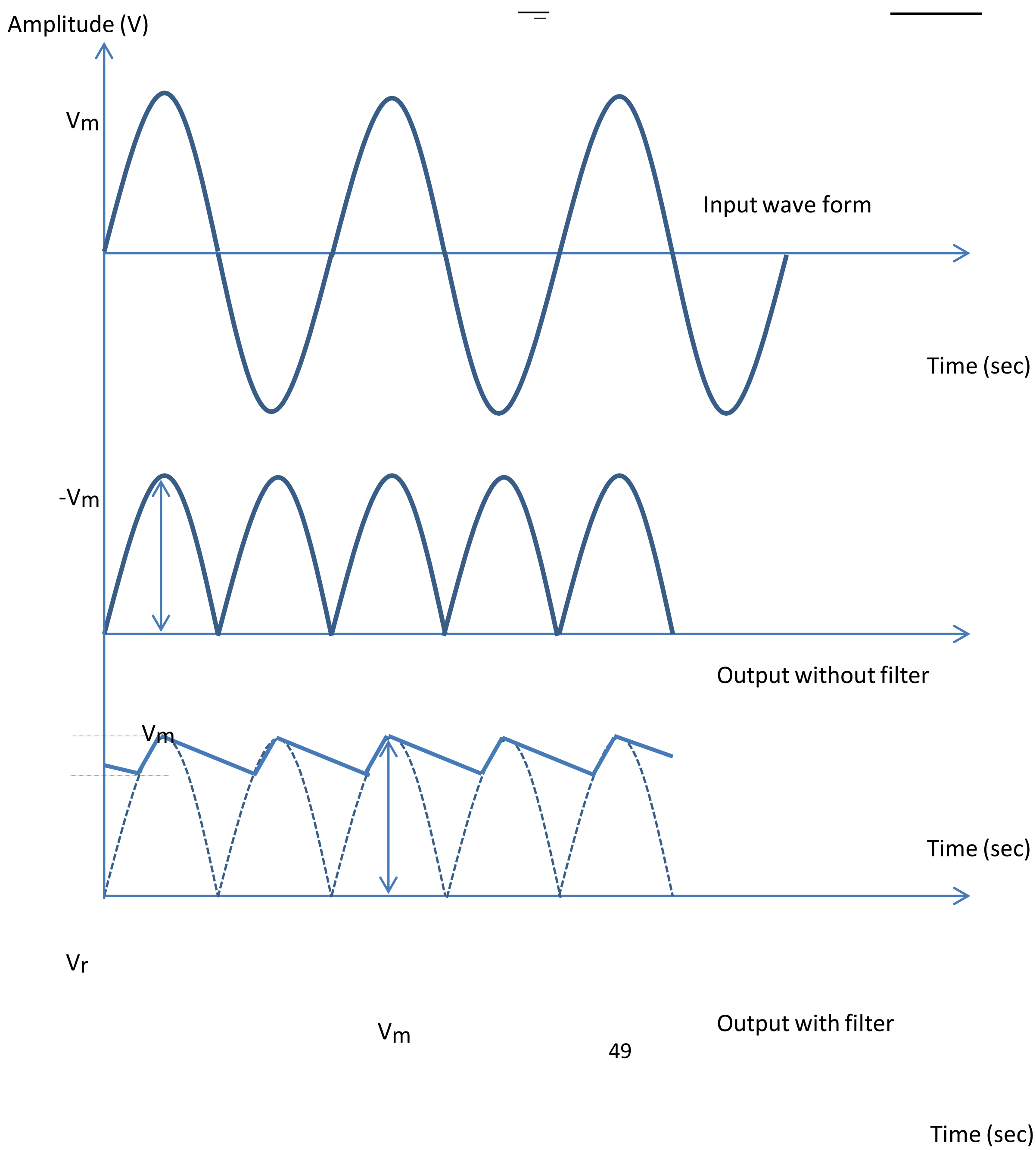
Using CRO : **V_{NL}=**

R_L (Ω)	V_L (V)	V_m (V)	$Vdc = \frac{2Vm}{\pi}$ (V)	$Vrms = \frac{Vm}{\sqrt{2}}$ (V)	$Vr(rms) = \sqrt{Vrms^2 - Vdc^2}$ (V)	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation $= \frac{V(NL)-V(L)}{VL}$

With Filter V_{NL}=

R_L (Ω)	V_L (V)	V_m (V)	V_r (V)	$Vdc = Vm - \frac{Vr}{2}$ (V)	$Vr(rms) = \frac{Vr}{4\sqrt{3}}$	R.F= $\frac{Vr(rms)}{Vdc}$	% Regulation $= \frac{V(NL)-V(L)}{VL}$

Model Graph:



Amplitude (V)
Vm
Input wave form
Time (sec)
-Vm
Output without filter
Vm
Time (sec)
Vr
Vm
Output with filter
Time (sec)

PRECAUTIONS:

1. The primary and secondary sides of the transformer should be carefully identified.
2. The polarities of the diode should be carefully identified.

Result: The input and output waveforms of Bridge wave rectifier is plotted and the ripple factor and regulation at 1100Ω are

Ripple factor without Filter =

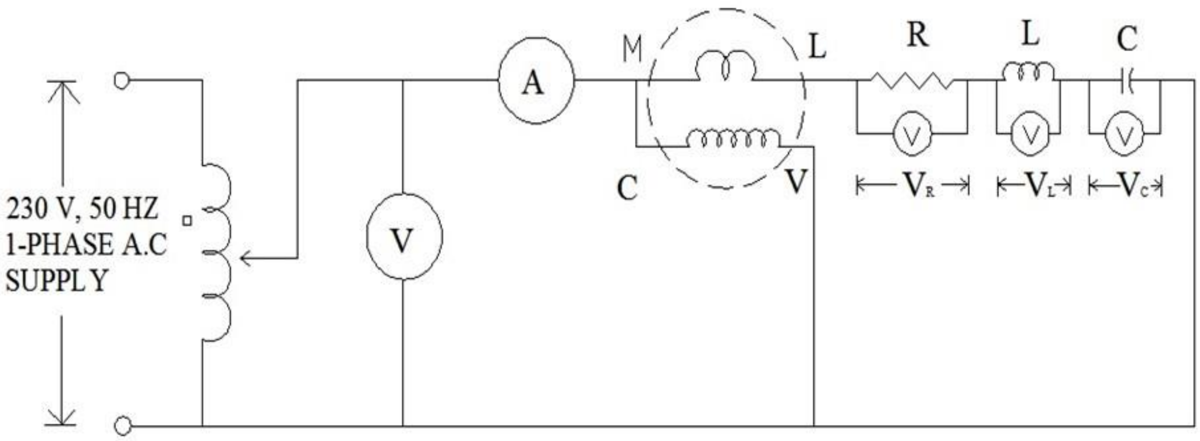
Ripple factor with Filter =

%Regulation=

SINGLE PHASE POWER MAESUREMENT

AIM: Measurement of single phase power in a R L C Circuit , obtain power factor and draw a Impedance triangle

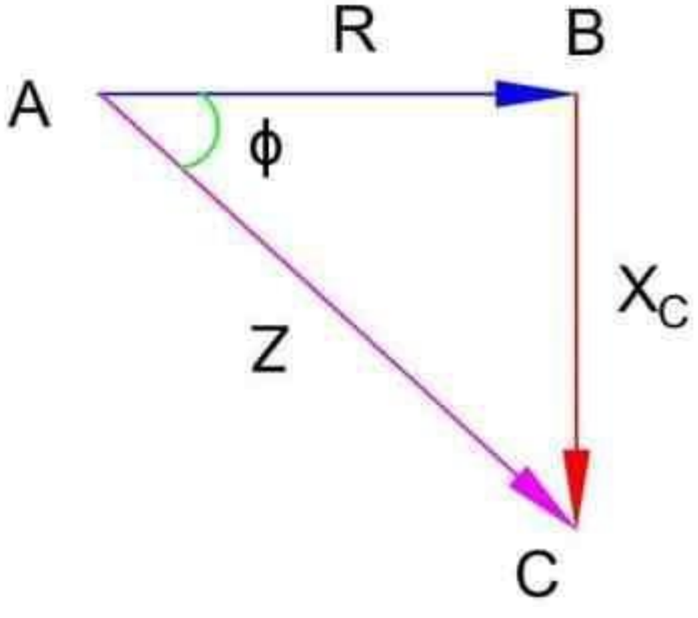
CIRCUIT DIAGRAM



I	V	W	VR	VL	VC	XL	XC	XC-XL	R	Z	COSφ

Z = $\sqrt{(R^2} + (XC - XL^2)$

COSφ = R/Z



MATERIALS REQUIRED:

SL NO	MATERIALS WITH SPECIFICATIONS	UNIT	QUANTITY
1	1mm^2 PVC insulated copper wire	cm	As per the given layout
2	Voltmeter	V	1
3	Ammeter	A	1
4	Resistor	R	1
	Inductor	Henry	1
	Capacitor	Farad	1
6	Auto Transformer		1
7	Wattmeter 10A,250 V LPF		1

TOOLS REQUIRED:

Screw driver, working board, pocketknife, pliers, steel rule, etc.

PRECAUTIONS:

- Bare portions of the conductor should not come out of the terminals
- All the connections should be tightened
- Switch should be connected always in phase wire
- Circuit should be checked by series test lamps

PROCEDURE:

- Draw the circuit diagram.
- Wire up the circuit as per the circuit diagram.
- Check the resistance capacitance and inductance of each component by using LCR and Multimeter.
- Switch ON the S P S T switch and note down the readings of V and I
- Calculate the voltage drop across the RLC circuit by using the noted readings. ➢ Check the circuit completely

RESULT:

The wiring has carried out successfully as per the given layout and calculations.